



The University of Azad Jammu and Kashmir

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Roll NO: 2024-SE-14

Semester:4th

Course Title: Database System

Course Code: CS-2204

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LAB 08

1. INTRODUCTION

This lab was performed to understand SQL JOIN operations using MySQL in XAMPP. Different types of joins including INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins were used to retrieve related data from multiple tables in the company schema.

2. OBJECTIVE

The objective of this lab was to practice SQL JOIN operations and retrieve related records from multiple database tables.

3. Create Database



4. Company schema tables created successfully.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ assignment	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ department	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ employee	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ project	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
4 tables	Sum	20	InnoDB	utf8mb4_general_ci	112.0 KiB	0 B

☐ Check all With selected:

Sample data inserted into all company schema tables.

`SELECT * FROM `assignment``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Sort by key: N

Extra options

				EmpID	ProjectID	WeeklyHours
☐	Edit	Copy	Delete	1	1	20
☐	Edit	Copy	Delete	2	3	15
☐	Edit	Copy	Delete	3	2	18
☐	Edit	Copy	Delete	6	1	25
☐	Edit	Copy	Delete	7	4	12

☐ Check all
 With selected: Edit Copy Delete Export

`SELECT * FROM `department``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Sort by key:

Extra options

				DeptID	DeptName	Location
☐	Edit	Copy	Delete	1	Engineering	Lahore
☐	Edit	Copy	Delete	2	HR	Karachi
☐	Edit	Copy	Delete	3	Marketing	Islamabad
☐	Edit	Copy	Delete	4	Finance	Lahore

☐ Check all
 With selected: Edit Copy Delete Export

SELECT * FROM `employee`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key:

Extra options

← T →						EmpID	EmpName	Salary	City	DeptID	ManagerID	
☐		Edit		Copy		Delete	1	Ali Khan	95000.00	Lahore	1	NULL
☐		Edit		Copy		Delete	2	Sara Ahmed	72000.00	Karachi	2	1
☐		Edit		Copy		Delete	3	Bilal Hussain	68000.00	Islamabad	3	1
☐		Edit		Copy		Delete	4	Maham Noor	88000.00	Lahore	2	1
☐		Edit		Copy		Delete	5	Usman Tariq	120000.00	Karachi	1	NULL
☐		Edit		Copy		Delete	6	Areeba Khan	85000.00	Lahore	1	5
☐		Edit		Copy		Delete	7	Daniyal Raza	76000.00	Islamabad	4	5

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

SELECT * FROM `project`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key:

Extra options

← T → ▼						ProjectID	ProjectName	DeptID	StartDate
☐	 Edit	 Copy	 Delete	1	Mobile App	1	2024-01-10		
☐	 Edit	 Copy	 Delete	2	Website Redesign	3	2024-03-15		
☐	 Edit	 Copy	 Delete	3	HR System	2	2023-11-20		
☐	 Edit	 Copy	 Delete	4	Finance Tracker	NULL	2024-02-05		

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

5. Part A Queries

TaskA1

- This query lists every employee with their department name and department location using INNER JOIN.

```
SELECT      e.EmpName,      d.DeptName,  
            d.Location  
FROM Employee e  
            INNER JOIN Department d  
ON e.DeptID = d.DeptID;
```

✓ Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)

-- Task A1 `SELECT e.EmpName, d.DeptName, d.Location FROM Employee e INNER JOIN Department d ON e.DeptID = d.DeptID;`

[Edit inline](#) [\[Edit \]](#) [\[Create PHP code \]](#)

☐ Show all | Number of rows: Filter rows:

Extra options

EmpName	DeptName	Location
Ali Khan	Engineering	Lahore
Sara Ahmed	HR	Karachi
Bilal Hussain	Marketing	Islamabad
Maham Noor	HR	Karachi
Usman Tariq	Engineering	Lahore
Areeba Khan	Engineering	Lahore
Daniyal Raza	Finance	Lahore

Output of Task A1 showing employees with their department names and locations.

TaskA2

- This query displays all employees including those whose department ID is NULL using LEFT JOIN.

```
SELECT      e.EmpName,      d.DeptName,  
            d.Location  
FROM Employee e  
LEFT JOIN Department d ON e.DeptID =  
d.DeptID;
```

✓ Showing rows 0 - 6 (7 total, Query took 0.0020 seconds.)

-- Task A2 `SELECT e.EmpName, d.DeptName, d.Location FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID;`

[Edit inline](#) [\[Edit \]](#) [\[Create PHP code \]](#)

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

EmpName	DeptName	Location
Ali Khan	Engineering	Lahore
Sara Ahmed	HR	Karachi
Bilal Hussain	Marketing	Islamabad
Maham Noor	HR	Karachi
Usman Tariq	Engineering	Lahore
Areeba Khan	Engineering	Lahore
Daniyal Raza	Finance	Lahore

Output of Task A2 showing all employees with department details.

TaskA3

- This query lists every department along with employee names. Departments without employees also appear in the result.

```

SELECT d.DeptName, e.EmpName
FROM Department d

        LEFT JOIN Employee e
        ON d.DeptID = e.DeptID;

```

✓ Showing rows 0 - 6 (7 total, Query took 0.0008 seconds.)

```
-- Task A3 SELECT d.DeptName, e.EmpName FROM Department d LEFT JOIN Employee e ON d.DeptID = e.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

DeptName	EmpName
Engineering	Ali Khan
Engineering	Usman Tariq
Engineering	Areeba Khan
HR	Sara Ahmed
HR	Maham Noor
Marketing	Bilal Hussain
Finance	Daniyal Raza

Output of Task A3 showing departments and their employees.

TaskA4

- This query displays every project with department name and location including projects without departments.

```

SELECT p.ProjectName, d.DeptName, d.Location FROM Project p

        LEFT JOIN Department d
        ON p.DeptID = d.DeptID;

```

✓ Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

```
-- Task A4 SELECT p.ProjectName, d.DeptName, d.Location FROM Project p LEFT JOIN Department d ON p.DeptID = d.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 ▼ | Filter rows:

Extra options

ProjectName	DeptName	Location
Mobile App	Engineering	Lahore
Website Redesign	Marketing	Islamabad
HR System	HR	Karachi
Finance Tracker	NULL	NULL

Output of Task A4 showing projects with department information.

TaskA5

- This query finds employees who are not assigned to any project using LEFT JOIN and IS NULL.

```
SELECT e.EmpName
FROM Employee e
LEFT JOIN Assignment a
ON e.EmpID = a.EmpID
WHERE a.ProjectID IS NULL;
```

✓ Showing rows 0 - 1 (2 total, Query took 0.0007 seconds.)

```
-- Task A5 SELECT e.EmpName FROM Employee e LEFT JOIN Assignment a ON e.EmpID = a.EmpID WHERE a.ProjectID IS NULL;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 ▾ | Filter rows: Search this table | Sort by key: None ▾

Extra options

EmpName
Maham Noor
Usman Tariq

Output of Task A5 showing employees without project assignments.

TaskA6

- This query lists projects that currently have no employee assignments.

```
SELECT p.ProjectName
      FROM Project p
LEFT JOIN Assignment a
ON p.ProjectID = a.ProjectID WHERE a.EmpID IS NULL;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

```
-- Task A6 SELECT p.ProjectName FROM Project p LEFT JOIN Assignment a ON p.ProjectID = a.ProjectID WHERE a.EmpID IS NULL;
```

[Edit inline] [Edit] [Create PHP code]

ProjectName

Query results operations

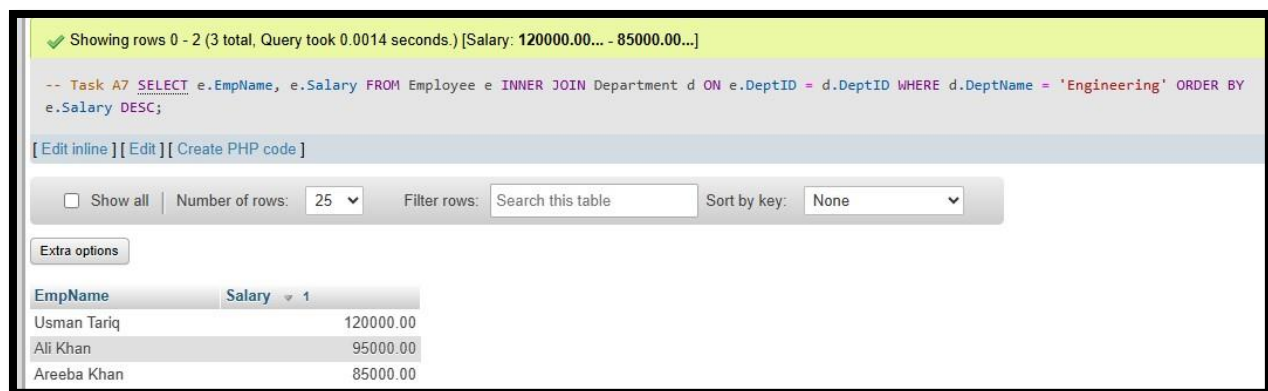
Output of Task A6 showing projects without assignments.

TaskA7

- This query displays employees in the Engineering department sorted by salary in descending order.

```
SELECT e.EmpName, e.Salary
      FROM Employee e
    INNER JOIN Department d
      ON e.DeptID = d.DeptID
     WHERE d.DeptName =
'Engineering'

      ORDER BY e.Salary DESC;
```



Showing rows 0 - 2 (3 total, Query took 0.0014 seconds.) [Salary: 120000.00... - 85000.00...]

```
-- Task A7 SELECT e.EmpName, e.Salary FROM Employee e INNER JOIN Department d ON e.DeptID = d.DeptID WHERE d.DeptName = 'Engineering' ORDER BY e.Salary DESC;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmpName	Salary
Usman Tariq	120000.00
Ali Khan	95000.00
Areeba Khan	85000.00

Output of Task A7 showing Engineering employees sorted by salary.

TaskA8

- This query lists employees working in departments located in Lahore.

```
SELECT e.EmpName, d.DeptName
      FROM Employee e
    INNER JOIN Department d
      ON e.DeptID = d.DeptID
 WHERE d.Location = 'Lahore';
```

Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

-- Task A8 `SELECT e.EmpName, d.DeptName FROM Employee e INNER JOIN Department d ON e.DeptID = d.DeptID WHERE d.Location = 'Lahore';`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmpName	DeptName
Ali Khan	Engineering
Usman Tariq	Engineering
Areeba Khan	Engineering
Daniyal Raza	Finance

Output of Task A8 showing employees in Lahore-based departments.

TaskA9

- This query displays every department with the total number of employees working in each department.

```
SELECT d.DeptName,  
COUNT(e.EmpID) AS  
EmployeeCount  
FROM Department d  
LEFT JOIN Employee e  
ON d.DeptID = e.DeptID  
GROUP BY d.DeptName;
```

✓ Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)

-- Task A9 `SELECT d.DeptName, COUNT(e.EmpID) AS EmployeeCount FROM Department d LEFT JOIN Employee e ON d.DeptID = e.DeptID GROUP BY d.DeptName;`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

DeptName	EmployeeCount
Engineering	3
Finance	1
HR	2
Marketing	1

Output of Task A9 showing employee count in each department.

TaskA10

- This query produces a FULL OUTER JOIN result of Employee and Department tables using UNION.

```

SELECT e.EmpName, d.DeptName
FROM Employee e
      LEFT JOIN Department d
ON e.DeptID = d.DeptID
UNION
SELECT e.EmpName, d.DeptName
FROM Employee e
      RIGHT JOIN Department d
ON e.DeptID = d.DeptID;

```

✓ Showing rows 0 - 6 (7 total, Query took 0.0015 seconds.)

-- Task A10 SELECT e.EmpName, d.DeptName FROM Employee e LEFT JOIN Department d ON e.DeptID = d.DeptID UNION SELECT e.EmpName, d.DeptName FROM Employee e RIGHT JOIN Department d ON e.DeptID = d.DeptID;

[Edit inline](#) [\[Edit \]](#) [\[Create PHP code \]](#)

☐ Show all | Number of rows: 25 ▼ | Filter rows: | Sort by key: None ▼

Extra options

EmpName	DeptName
Ali Khan	Engineering
Sara Ahmed	HR
Bilal Hussain	Marketing
Maham Noor	HR
Usman Tariq	Engineering
Areeba Khan	Engineering
Daniyal Raza	Finance

Output of Task A10 showing FULL OUTER JOIN using UNION.

6. Conclusion

In this lab, different SQL JOIN operations were performed successfully using MySQL in

XAMPP. INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins helped in retrieving related information from multiple tables. The lab improved understanding of relational database concepts and SQL query writing.

*****THE END*****